



Dietary partitioning between sympatric species of spiny lobster in a coral reef system

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ABSTRACT.—Throughout the Wider Caribbean Region, the spotted spiny lobster, *Panulirus guttatus* (Latreille, 1804), dwells and forages exclusively in coral reef habitat after post-larval settlement, whereas its sympatric species, the Caribbean spiny lobster, *Panulirus argus* (Latreille, 1804), undergoes ontogenetic habitat shifts, with smaller individuals dwelling in reef lagoon habitats and larger individuals dwelling in reef habitats, but exhibiting broad foraging ranges. Stomach content, and carbon and nitrogen stable isotopes analyses were used to examine dietary partitioning between co-occurring *P. guttatus* and reef-dwelling *P. argus* in a coral reef system off Mexico. Both species fed on a wide variety of invertebrates, but mollusks contributed more to the diet of *P. guttatus* and crustaceans to the diet of *P. argus*. The two species had different ranges of isotopic composition and each filled a unique trophic position within the local food web. A Bayesian mixing model revealed a distinct contribution of different food sources to the diet of these lobsters. Reef herbivores were the primary food source for *P. argus* and an important food source for *P. guttatus*, but the latter also consumed more carnivores, resulting in a higher trophic position relative to *P. argus*. The isotopic niche space, delineated by means of bivariate ellipses, exhibited no overlap between species, and *P. argus* exhibited a significantly larger niche area than *P. guttatus*. These results suggest that, in addition to a differential use of habitat resources, a differential use of food resources facilitates the local coexistence of these two congeners.

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Competition occurs when two or more species use the same resource that is in limited supply, and has been considered an important evolutionary force leading to niche separation or specialization (Pianka 1978). Basic principles of niche theory suggest that complete niche overlap between sympatric species is evolutionarily impossible (Gause 1934, Hardin 1960); therefore, to coexist, species must differ in some way (i.e., reduced niche overlap) (Chesson 2000). Differential use of vital resources (e.g., food or habitat) may facilitate the local coexistence of ecologically-similar or closely-related species (Kneitel and Chase 2004, Lozano-Álvarez et al. 2007).

Spiny lobsters (Crustacea: Palinuridae) are large, nocturnal, crevice-dwelling crustaceans that represent important fishing resources in all tropical and subtropical seas and occur in coral reef habitats throughout the world (Phillips 2000). The Caribbean spiny lobster, *Panulirus argus* (Latreille, 1804), and the spotted spiny lobster, *Panulirus guttatus* (Latreille, 1804), are sympatric across the Wider Caribbean Region (Holthuis 1991). *Panulirus guttatus* is a habitat specialist that dwells exclusively on the coral reef habitat throughout its benthic life, whereas *P. argus* undergoes ontogenetic habitat shifts during its benthic life (reviewed in Briones-Fourzán 2014). The post-larvae of *P. argus* settle in coastal submerged vegetated habitats of shallow embayments and reef lagoons, where the small juveniles dwell for a few months before shifting to crevice-type structures that provide shelter in shallow waters. Sub-adults migrate to the coral reefs, which constitute the main adult habitat. Hence, sub-adult and adult *P. argus* co-occur with *P. guttatus* in the coral reef habitat (Lozano-Álvarez et al. 2007).

Previous studies have revealed a differential use of habitat and shelter resources in coral reefs between these two congeneric lobsters (Sharp et al. 1997, Acosta and Robertson 2003, Lozano-Álvarez et al. 2007, Briones-Fourzán 2014). In contrast, gut content analyses have suggested similar diets in terms of gross food items (i.e., crustaceans, molluscs, etc.) (Colinas-Sánchez and Briones-Fourzán 1990, Cox et al. 1997, Butler and Kintzing 2016). However, competition for food resources between these two species is believed to be low because *P. guttatus* exhibit a limited home range and tend to forage close to their reef dens (Sharp et al. 1997, Lozano-Álvarez et al. 2002, Wynne and Côté 2007), whereas reef-dwelling *P. argus* are highly mobile and tend to forage in different kinds of habitats (e.g., reef, rubble, seagrass) over 10s to 100s of meters (Herrnkind et al. 1975, Cox et al. 1997, Acosta 2002, Bertelsen 2013).

In some spiny lobster species, gut content analyses suggest changes in diet with size [e.g., *Panulirus cygnus* George, 1962: Edgar 1990; *Palinurus elephas* (Fabricius, 1787): Goñi et al. 2001; *Jasus lalandii* (H. Milne-Edwards, 1837): Mayfield et al. 2001], although diet changes appear to be more related to local availability of specific prey than to lobster size in highly mobile species such as *P. argus* (Herrnkind et al. 1975, Cox et al. 1997, Briones-Fourzán et al. 2003), *Panulirus interruptus* (Randall, 1840) (Castañeda-Fernández-de-Lara et al. 2005), and *P. cygnus* (Waddington et al. 2008, MacArthur et al. 2011). However, gut content analysis provides only a snapshot of what an animal has consumed up to a few hours prior to capture. For spiny lobsters in particular, this technique may limit the assessment of true diet composition and level of competition for dietary resources due to the opportunistic feeding exhibited by these animals, the high level of trituration of the prey by their mandibles and gastric mill, and their rapid digestion and evacuation rates (Colinas-Sánchez and Briones-Fourzán 1990, Edgar 1990).

Stable isotope analysis (SIA), particularly the measurement of $^{13}\text{C}/^{12}\text{C}$ and $^{15}\text{N}/^{14}\text{N}$ in soft tissues of heterotrophs, has been extensively used to examine trophic relationships in a variety of marine systems. The isotopic composition of consumers' tissues typically resemble that of their food sources, although there is discrimination against the heavy isotopes of carbon and nitrogen (^{13}C and ^{15}N) during respiration and nitrogen metabolism that leads to a predictable enrichment in the heavy isotopes with increasing trophic level. Values of $\delta^{13}\text{C}$ reflect the dominant sources of primary production supporting an animal's diet, whereas values of $\delta^{15}\text{N}$ are useful for estimating trophic position due to its higher level of trophic discrimination

(Bearhop et al. 2004). SIA provides a time-integrated view of the trophic relationships of an individual over time (up to several weeks or months, depending on the type of tissue). For instance, in the red rock lobster *Jasus edwardsii* (Hutton, 1875), the estimated half-life for muscle tissue is 147 d (Suring and Wing 2009).

Furthermore, by allowing quantification of the trophic components of the niche space or “isotopic niche,” SIA has become a valuable tool for assessing resource use and niche variation at different ecological scales (Bearhop et al. 2004, Layman et al. 2007, Newsome et al. 2012). SIA has been especially useful in the study of resource partitioning between sympatric congeneric species [e.g., mysids: Lehtiniemi et al. (2009); krill: Agersted et al. (2014); fishes: Ho et al. (2007), Dromard et al. (2013), Knickle and Rose (2014); rorquals: Ryan et al. (2013)], which is the focal question of our study.

The main objective of the present study was to assess dietary differences between *P. guttatus* and *P. argus* lobsters co-occurring in a coral reef system by using two complementary approaches: gut content analysis to determine general dietary habits and SIA to infer diet composition and niche metrics.

MATERIALS AND METHODS

STUDY AREA.—The study was conducted in the Puerto Morelos coral reef system, located off the northeast coast of the Yucatán Peninsula, Mexico (20°51'N, 86°53'W). This coral reef system includes an extended fringing reef and a shallow (<5 m in depth), relatively narrow (<1500 m in width) reef lagoon extending between the reef and the coastline. The reef lagoon contains extensive seagrass meadows interspersed with patches of sand and calcareous pavement (Ruiz-Rentería et al. 1998). Diverse macroalgae abound in both the coral reef and the lagoon (Collado-Vides et al. 1998). The reef has well-developed back-reef and crest zones, but a relatively flat fore-reef zone that descends gradually to 20–25 m into an extensive sand platform (Jordán-Dahlgren 1989). This reef configuration results in greater coexistence of *P. argus* and *P. guttatus* on the back-reef zone, where both species also tend to have similar mean sizes (Lozano-Álvarez et al. 2007). Therefore, lobsters were collected from the back-reef zone.

SAMPLE COLLECTION AND GUT EXAMINATION.—We collected 20 *P. argus* and 23 *P. guttatus* from the back-reef zone via scuba diving. The lobsters were collected by hand between 20:00 and 23:00 hrs, when lobsters leave their dens to forage, and were transported to the laboratory within 1 hr of collection. The lobsters were measured (carapace length, CL) to the nearest 0.1 mm with Vernier calipers, and their guts were dissected and individually preserved in 70% ethanol until analysis. Gut contents were analyzed following Briones-Fourzán et al. (2003). Due to their fragmentation or partial digestion, contents were grouped in food categories corresponding to gross taxa. For each food category, we determined percent frequency of occurrence ($\%F$ = number of stomachs containing a given food category/total number of stomachs examined $\times$ 100), percent volume ($\%V$ = percent contribution to the total volume of stomach contents), and an index of relative importance [$IRI = (\%F \times \%V)/100$] (Briones-Fourzán et al. 2003, Castañeda-Fernández-de-Lara et al. 2005). Using the $\%V$ of food categories, we computed Horn's index of overlap (Horn 1966) to examine a potential overlap in diet between both species. Horn's index ranges from 0 (no

overlap) to 1 (complete overlap). Diets of different species can be considered as substantially overlapping when the index value is ≥ 0.60 (Zaret and Rand 1971).

Potential prey for lobsters were collected from the reef habitat and also from the adjacent lagoon habitat, given that reef-dwelling *P. argus* can forage 10s to 100s of meters away from the reef (Bertelsen 2013). Several of the most common species of macroalgae and the seagrass *Thalassia testudinum* Bank ex König were collected and analyzed in duplicate to characterize the isotopic composition of primary producers.

STABLE ISOTOPE ANALYSES (SIA).—For SIA, subsamples of tail muscle from 14 *P. guttatus* (size range: 41.9–76.0 mm CL) and 12 *P. argus* (42.8–75.6 mm CL) were rinsed with deionized water, dried completely at 60 °C, then ground to a fine powder with an agate mortar and pestle. Sample preparation of biota from reef and lagoon habitats was similar. Macroalgal samples containing calcium carbonate were treated with 1 M HCl to remove inorganic carbonate (Boutton 1991). Seagrass leaves were scraped to remove epiphytes and also acidified as mentioned above. Fine powdered tissue was transferred into tin capsules. For all samples, $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ were determined using a PDZ Europa ANCA-GSL elemental analyzer interfaced to a PDZ Europa 20-20 isotope ratio mass spectrometer at the University of California Davis Stable Isotope Facility. Carbon and nitrogen ratios were expressed in delta (δ) notation, in units of per mil (‰): $\delta = 1000 \times (R_{\text{sample}} - R_{\text{standard}}) / R_{\text{standard}}$, where R is the ratio of the heavier, rarer isotopes (^{13}C and ^{15}N) over the lighter, more common isotopes (^{12}C or ^{14}N). Delta values are reported relative to the international standards of Vienna Pee-Dee Belemnite carbon and atmospheric nitrogen. Analytical error is reported as recommended by Jardine and Cunjak (2005); samples were interspersed with 50 replicates of a commercial laboratory analytical standard (Nylon 5); the mean accuracy was -27.72 (SD 0.041) for $\delta^{13}\text{C}$ and -10.31 (SD 0.112) for $\delta^{15}\text{N}$, and the average precision was 0.1‰ for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ across runs.

The potential prey items collected from each habitat (reef or lagoon) were categorized into three trophic groups—herbivores, omnivores, and carnivores—based on published diet data (e.g., Lozet and Petron 1977, Colin 1978, Abbott 1986, Hendler et al. 1995). These resulted in six potential food sources (three trophic groups $\times$ two habitats). The likely contribution of these food sources to the lobster diets was determined with a Bayesian isotope mixing model (R package SIAR, Parnell et al. 2010) using the mean enrichment values [3.2 (SD 0.31) for C and 2.57 (SD 0.65) for N] calculated for the western rock lobster, *P. cygnus* (Waddington and MacArthur 2008). The isotopic mixing models are the most common approach to estimate the proportional contribution of different food sources (Post 2002). However, applying the mixing model using many sources with potentially similar isotopic values might prevent finding the right solution of source proportions; that is, the mixing model might not be able to discriminate between groups (Phillips and Gregg 2003). Therefore, we used a randomization test to assess whether the individual trophic groups (i.e., herbivores, omnivores, carnivores) were statistically similar between the two habitats sampled (reef and lagoon) by examining the k-mean total sum of squares (KMTSS), as this incorporated variation in both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ simultaneously and could thus test (at an α level of 0.05) whether the composition of the two isotopes in each group was random. For each trophic group, comparisons of the isotope pairs from both habitats were randomized and re-assigned 10,000 times for each habitat. A randomization test was also used to test for differences between isotopic composition of the two

lobster samples. This test was chosen as it has been successfully used to distinguish stable isotope clusters (Hooker et al. 2001).

The isotopic niche width for each lobster species was determined based on the isotopic dispersion of samples within a two-dimensional ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) space. Bivariate ellipses, estimated as standard ellipse area (SEA), were used as a measure of the total trophic niche area for each lobster species. The Bayesian standard ellipses (SEAc), corrected for small sample sizes by 2 standard deviations ($n - 1$ for each axis), were also estimated and then plotted using the SIBER routine for the SIAR package in R (Jackson et al. 2011). Distinction between the two SEAc was inferred by posterior probabilities derived by Bayesian inference based on 100,000 Markov Chain Monte Carlo draws. The trophic niche overlap was calculated as the proportion of SEAc overlapping between lobster species. Finally, to assess potential size-related changes in trophic level (e.g., Waddington et al. 2008), the relationship between lobster size and $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values was evaluated for each species using general linear regression analyses.

RESULTS

The mean CL of lobsters used for gut content analyses was slightly larger for *P. argus* [62.1 (SD 11.7) mm CL] than for *P. guttatus* [56.2 (SD 8.6) mm CL], but the difference was not significant (Student's *t*-test: $t_{38} = 1.507$, $P = 0.07$). Two *P. argus* and three *P. guttatus* had empty stomachs. Both species fed on a wide variety of invertebrates, especially crustaceans (mostly brachyuran and anomuran crabs, but also isopods) and mollusks (mostly gastropods, but also occasionally bivalves and chitons), as well as echinoderms (sea urchins, brittle stars), polychaetes, sponges, and some plant material (macroalgae) and coralline algae (Table 1). *Panulirus argus* also consumed foraminifers, suggesting foraging over soft substrates, and fish, suggesting some scavenging. Crustaceans contributed more in volume to the diet of *P. argus* (40%) than to the diet of *P. guttatus* (18.3%), whereas mollusks contributed more in volume to the diet of *P. guttatus* (41%) compared with *P. argus* (16.3%). This pattern was reflected in the contrasting IRIs of crustaceans and mollusks between spiny lobster species (Table 1). Nevertheless, the diet of both species showed significant overlap (Horn's index: 0.73) in terms of gross taxa.

Among the primary producers analyzed, the seagrass *T. testudinum* was relatively enriched in ^{13}C ($-8.99\text{‰} \pm 0.032\text{‰}$) and ^{15}N ($3.58\text{‰} \pm 0.056\text{‰}$) compared with the seven species of macroalgae, which ranged from -19‰ to -14‰ for ^{13}C , and from 0.75‰ to 2.35‰ for ^{15}N (Table 2). These isotopic values for *T. testudinum* and macroalgae are consistent with previous studies conducted in Florida (Zieman et al. 1984, Behringer and Butler 2006). Relative to seagrass, the mean carbon isotope composition of all consumers, including the two lobster species, was depleted in ^{13}C (range: -16.55‰ to -8.91‰) (Table 2).

In general, the trophic groups from the reef and the lagoon showed dissimilar $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ signatures (Fig. 1). Indeed, the randomization tests showed that the isotope compositions of herbivores and omnivores differed significantly between the reef and the lagoon ($P = 0.037$ and 0.007 , respectively). This was not the case for carnivores, but they were kept separated by habitat because the carnivore species differed between the two habitats. Summary of mean $\delta^{13}\text{C}$ and $\delta^{15}\text{N} \pm 1$ SD values of the trophic groups are shown in Table 2.

Table 1. Summary of stomach content analyses for *Panulirus argus* ($n = 18$) and *Panulirus guttatus* ($n = 20$) from the back reef habitat. %V: percent volume; %F: percent frequency; IRI: Index of Relative Importance = $(\%F \times \%V)/100$.

Food categories	<i>Panulirus argus</i>			<i>Panulirus guttatus</i>		
	%V	%F	IRI	%V	%F	IRI
Crustaceans	40.0	100.0	40.08	15.5	35.0	5.43
Mollusks	16.3	88.9	14.45	33.3	65.0	21.61
Echinoderms	9.5	38.9	3.70	3.8	20.0	0.75
Polychaetes	1.8	38.9	0.70	12.0	20.0	2.40
Sponges	1.4	22.2	0.31	6.0	20.0	1.20
Plant matter	2.1	66.7	1.37	19.8	45.0	8.89
Coralline algae	0.4	33.3	0.15	5.0	5.0	0.25
Foraminiferans	0.6	77.8	0.47	0.0	0.0	0.00
Gorgonians	0.1	22.2	0.01	0.0	0.0	0.00
Fish	0.3	11.1	0.04	0.0	0.0	0.00
Sediments	8.1	88.9	7.22	0.0	0.0	0.00
Soft matter	19.3	100.0	19.34	4.8	15.0	0.71

Mean size of lobsters used in SIA did not differ significantly between species [*P. guttatus*: 59.4 (SD 5.6) mm CL; *P. argus*: 60.1 (SD 7.1) mm CL; $t_{25} = 0.157$, $P = 0.876$]. Lobster size did not show a significant relationship with either $\delta^{13}\text{C}$ or $\delta^{15}\text{N}$ values for either species ($P > 0.05$ in all cases). The k-means test statistic indicated that the isotope compositions of the two lobster samples differed significantly ($P = 0.005$) (Table 3). The estimation of the likely contribution of food sources to the diet of lobsters showed differences between species (Fig. 2). Thus, *P. guttatus* fed mostly on reef herbivores and consumed a higher proportion of carnivores than *P. argus*, for which

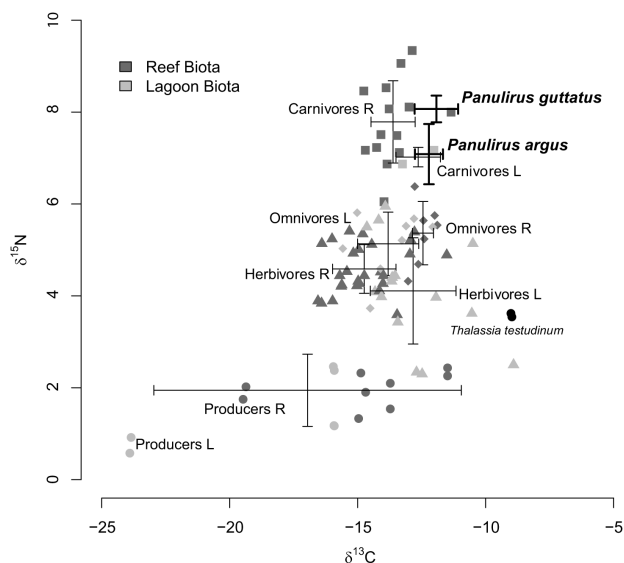


Figure 1. Bi-plot of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ isotopic values mean (SD) of spiny lobsters *Panulirus guttatus* and reef-dwelling *Panulirus argus* and their potential food sources—herbivores (triangles); omnivores (diamonds); carnivores (squares)—collected in the coral reef (dark gray symbols) and lagoon habitats (light gray symbols). The group of Producers (circles) only includes macroalgae and is shown only as a reference of primary productivity. The seagrass *Thalassia testudinum* is shown separately.

Table 2. Species included in the study, separated by habitat (lagoon and reef) and showing the sample size (n) and the mean and standard deviation (SD, in parentheses) of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values (‰) per species. Producers were not included in the mixing model analysis but are considered as a reference of primary productivity. Herbivores, omnivores, and carnivores from each habitat were included in the mixing model analysis, hence their mean (SD) of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values (‰) are shown in bold. See Online Appendix 1 for species authorities.

Trophic group/species	Reef habitat			Lagoon habitat		
	n	$\delta^{13}\text{C}$	$\delta^{15}\text{N}$	n	$\delta^{13}\text{C}$	$\delta^{15}\text{N}$
Producers						
<i>Thalassia testudinum</i>				2	-8.99 (0.03)	3.58 (0.06)
<i>Caulerpa lanuginosa</i>				2	-23.87 (0.04)	0.75 (0.24)
<i>Halimeda incrassata</i>				2	-15.92 (0.02)	1.18 (0.00)
<i>Halimeda discoidea</i>	2	-19.42 (0.09)	1.88 (0.19)	2	-15.93 (0.03)	2.42 (0.06)
<i>Caulerpa cupressoides</i>	2	-11.49 (0.00)	2.35 (0.12)			
<i>Dyctiota dichotoma</i>	2	-14.20 (0.69)	1.72 (0.25)			
<i>Chondria littoralis</i>	2	-14.29 (0.82)	2.21 (0.16)			
<i>Padina</i> sp.	1	-14.96	1.33			
Herbivores	28	-14.82 (1.29)	4.55 (0.55)	15	-12.83 (1.67)	4.10 (1.16)
<i>Phyllonotus pomum</i>	1	-16.96	3.64			
<i>Columbella mercatoria</i>	3	-16.31 (0.30)	3.87 (0.03)			
<i>Pecten</i> sp.	2	-15.33 (0.53)	4.72 (0.41)			
<i>Paguristes cadenati</i>	2	-14.99 (0.05)	4.24 (0.04)			
<i>Paguristes tortugae</i>	5	-15.66 (0.52)	5.05 (0.34)			
<i>Echinometra viridis</i>	2	-14.62 (0.25)	5.23 (0.16)			
<i>Mithraculus sculptus</i>	3	-12.89 (0.65)	5.17 (0.46)	1	-13.92	5.95
<i>Calcinus tibicen</i>	3	-14.13 (0.54)	4.35 (0.12)	3	-10.54 (1.58)	3.91 (0.30)
<i>Tripneustes</i> sp.	1	-11.52	4.89	2	-13.54 (0.16)	3.91 (0.69)
<i>Lithopoma tectum</i>	3	-14.91 (0.74)	4.22 (0.11)	2	-12.59 (0.16)	2.32 (0.03)
<i>Paguristes puncticeps</i>	3	-14.57 (0.92)	4.39 (0.16)	3	-13.76 (0.28)	4.25 (0.24)
<i>Cerithium eburneum</i>				1	-8.91	2.50
<i>Diadema</i> sp.				1	-10.50	5.14
<i>Petrolisthes galathinus</i>				2	-14.41 (0.33)	5.57 (0.11)
Omnivores	7	-12.45 (0.45)	5.37 (0.69)	8	-13.81 (1.20)	5.13 (0.69)
Alpheidae	1	-11.99	5.75			
<i>Tonicia schrammi</i>	1	-13.03	4.32			
<i>Cerithium eburneum</i>	3	-12.48 (0.13)	5.19 (0.48)			
<i>Macrocypraea zebra</i>	2	-12.33 (0.62)	5.96 (0.59)			
<i>Arca zebra</i>				1	-15.02	5.81
<i>Barbatia candida</i>				1	-15.58	5.03
<i>Damithrax hispidus</i>				3	-13.06 (0.24)	5.47 (0.24)
<i>Mithraculus forceps</i>				3	-13.56 (1.31)	4.61 (0.89)
Carnivores	14	-13.62 (0.89)	7.79 (0.89)	2	-12.63 (0.87)	6.87 (0.21)
<i>Fisurella</i> sp.	2	-13.18 (0.28)	7.62 (0.70)			
<i>Cyphoma gibbosum</i>	2	-13.72 (0.35)	6.77 (1.02)			
<i>Cassis tuberosa</i>	1	-11.35	8.00			
<i>Monoplex pilearis</i>	1	-14.09	7.51			
<i>Mitra barbadensis</i>	1	-13.31	9.06			
<i>Portunus</i> sp.	2	-14.05 (0.29)	7.05 (0.25)			
<i>Cinetorhynchus rigens</i>	1	-14.70	7.17			
<i>Eucidaris tribuloides</i>	3	-13.83 (0.77)	8.60 (0.53)			
<i>Phyllonotus pomum</i>				1	-12.02	7.17
<i>Leucozonia nassa</i>				1	-13.25	6.87

Table 3. Mean and associated standard deviation (in parentheses) of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ signatures and the estimated standard ellipse trophic area (SEA) and corrected standard ellipse trophic area (SEAc) of *Panulirus guttatus* and *P. argus* from a back reef habitat.

Lobster group	<i>n</i>	$\delta^{13}\text{C}$	$\delta^{15}\text{N}$	SEA	SEAc
<i>Panulirus guttatus</i>	14	-11.92 (0.85)	8.07 (0.29)	0.709	0.768
<i>Panulirus argus</i>	12	-12.21 (0.55)	7.08 (0.65)	0.900	0.990

reef herbivores constituted by far the primary food source. Omnivores in general constituted a secondary food source for both lobster species.

Each lobster species occupied a distinct isotopic niche space (Table 3). Isotopic niche area estimated by SEAc was significantly greater for *P. argus* than for *P. guttatus* ($P < 0.001$) (Table 3, Fig. 3). Moreover, there was no overlapping of SEAc between both lobster groups (Fig. 3), strongly suggesting no dietary overlap in terms of food sources (trophic levels). This result, in conjunction with the greater occurrence of carnivores in the diet of *P. guttatus*, further suggests that this obligate reef-dweller species exhibits a higher trophic position relative to *P. argus*.

DISCUSSION

The present study is the first using stable isotope and gut content analyses to examine differences in diet between spiny lobster species that co-occur in coral reef habitats. As noted by Orians and Horn (1969), measuring food overlap between co-occurring species is complex because two individuals in the same habitat may find different foods by foraging in different ways or obtain the same foods in slightly different habitats. Indeed, despite a significant diet overlap (Horn's index) between *P. guttatus* and *P. argus* in terms of gross taxa based on stomach content analyses, results from SIA suggested that food competition between both species does not occur at the local scale in the reef system studied, as these species occupied distinct isotopic niches. The inconsistency between results from both types of analyses likely reflects the lack of resolution in the identification of the stomach contents that resulted in pooling food items in gross taxa. For example, Espinosa et al. (1990) found a 77.1% similarity in the diet of *P. argus* collected in seagrass habitats and reef habitats from

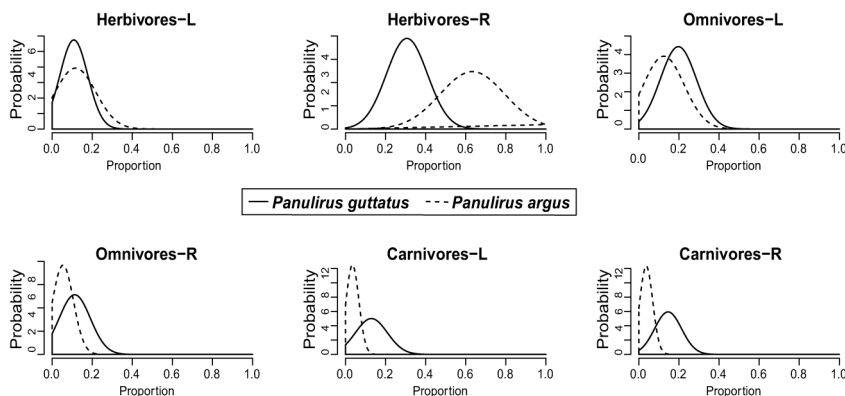


Figure 2. Estimated density proportions of the contribution of the six food sources to the diet of each of the two lobster species (*Panulirus guttatus* and reef-dwelling *Panulirus argus*), as determined by the Bayesian mixing model.

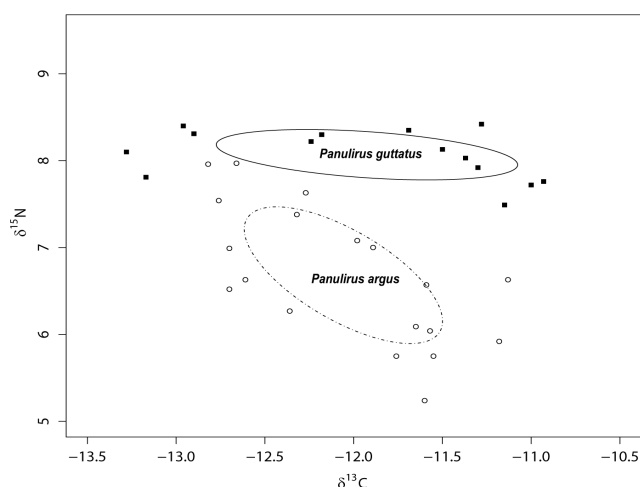


Figure 3. Bi-plot of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ isotopic values showing trophic niche for the two lobster species, *Panulirus guttatus* (black squares) and reef-dwelling *Panulirus argus* (open circles). Enclosed areas represent the corrected standard ellipse trophic area (SEAc) occupied by each species, which are described by the covariance in both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ and incorporate data points that better describe the trophic niche for each lobster species.

Cuba based on gross taxa, but a similarity of 13%–25% (depending on the season) when the food items were identified to species level. Moreover, although crustaceans and mollusks are important prey for *P. argus* and *P. guttatus*, different studies have found substantial differences in the relative contribution of these taxa to the diet of each lobster species. Thus, we found a much higher contribution of crustaceans to the diet of reef-dwelling *P. argus* compared with *P. guttatus*, and the opposite pattern for mollusks. In contrast, in the Florida Keys, mollusks accounted for 74.6% and arthropods for 11.7% of stomach contents in *P. argus* (Cox et al. 1997), whereas crabs accounted for 48% and mollusks for 22% of stomach contents in *P. guttatus* (Butler and Kintzing 2016). These discrepancies possibly reflect differences in local availability of different types of prey at the time of sampling, given the opportunistic feeding of both species of spiny lobster. For example, in laboratory feeding trials, *P. guttatus* did not show preference for crabs, gastropods, or sea urchins (Butler and Kintzing 2016), and in the US Virgin Islands, individuals of *P. argus* dwelling in a reef habitat fed more on crustaceans, whereas those dwelling in a lagoon habitat fed more on gastropods (Herrnkind et al. 1975).

The estimates of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ obtained for reef-dwelling *P. argus* in the present study were consistent with isotopic values previously estimated for *P. argus* from different habitats in the Florida Keys (Table 3) (Behringer and Butler 2006), but to our knowledge, there are no previous estimates for *P. guttatus*. In the Florida Keys, the isotopic signatures of consumers strongly suggested that macroalgae were a more important source for secondary production than seagrass (Behringer and Butler 2006), and we obtained a similar result in our study. Although the isotopic composition of macroalgae is particularly variable, reflecting the diverse energy pathways into the benthic community, consumers may reduce some of this variability by feeding upon many different plants or in different patches (Peterson 1999).

Panulirus guttatus showed a narrow niche in $\delta^{15}\text{N}$, which might result from extent of prey specialization, but a wide spread in $\delta^{13}\text{C}$ values, suggesting potential differences in primary carbon sources at different spatial scales across the reef habitat. Individuals of *P. guttatus* forage on a variety of benthic substrates on patch reefs, including reef rubble and macroalgae (Butler and Kintzing 2016), but tend to remain in the same reef areas or patches for years at a time and do not tend to venture far from the reef to forage (Negrete-Soto et al. 2002, Robertson and Butler 2009). Small individuals (<45 mm CL) feed in recesses close to their reef dens (Wynne and Côté 2007), but most of the *P. guttatus* that we sampled were >45 mm CL and many were observed feeding over exposed macroalgal patches on the reef, and even along the back reef-seagrass interface, suggesting that they search for small prey that use the reef vegetation as a refuge. If so, then the disparate $\delta^{13}\text{C}$ values could reflect the dominant macroalgae in the particular foraging area of each individual. Foraging in the back-reef seagrass interface could also partially explain the relatively high contribution of lagoon omnivores to the diet of *P. guttatus*. Alternatively, the contribution of lagoon omnivores to the diet of *P. guttatus* might reflect a lack of resolution of the mixing model, given that the food sources were composed of distinct prey items (i.e., the same prey species were not always collected from both habitats). In contrast with *P. guttatus*, reef-dwelling *P. argus* lobsters showed a broader isotopic niche area for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$, which is consistent with a more flexible foraging behavior involving the use of other foraging habitats in addition to the coral reef (e.g., rubble, adjacent seagrass meadows) (Herrnkind et al. 1975, Cox et al. 1997, Briones-Fourzán et al. 2003, Bertelsen 2013), thus reducing the potential for competition for food resources with *P. guttatus*.

Waddington et al. (2008) found that larger western rock lobsters, *P. cygnus*, were more carnivorous than smaller conspecifics. In contrast, in Tasmanian red rock lobsters, *J. edwardsii*, Guest et al. (2009) found no relationship between lobster size and values of either $\delta^{15}\text{N}$ or $\delta^{13}\text{C}$, possibly because changes in diet over the size range of the individuals they examined involved a change in the proportion of existing food sources rather than a replacement of food types (Guest et al. 2009). Similarly, we did not find a relationship between lobster size and values of either $\delta^{15}\text{N}$ or $\delta^{13}\text{C}$ for the two spiny lobster species, suggesting that, for each species, food types were similar for lobsters over the size ranges that we examined. It would be interesting to explore potential changes in isotopic signatures over a broader size range in *P. argus*, which reaches larger sizes (>100 mm CL) than the ones we sampled.

Although spiny lobsters are considered to be generalist feeders, the present study revealed that reef-dwelling *P. argus* and *P. guttatus* integrate unique sources of carbon and nitrogen from their environment, resulting in minimal dietary overlap. This finding is consistent with the hypothesis that interspecific dietary competition between *P. guttatus* and *P. argus* does not occur at the local scale in the Puerto Morelos reef system. SIA has also revealed differences in diet in other coexisting species of marine organisms typically known as generalist feeders. For example, SIA revealed that among three species of sea urchins co-occurring in rocky reefs in SW Australia, one was almost exclusively herbivorous, whereas the other two had an omnivorous diet and exhibited a higher trophic position consistent with the consumption of a wider variety of food items (Vanderklift et al. 2006). Dietary separation as assessed with SIA was also found to preclude exploitative competition among sympatric reef fish species off Taiwan (Ho et al. 2007) and the Caribbean Sea (Dromard et al. 2013),

and between co-occurring species of gobies off China (Guo et al. 2014) and sea urchins off Ecuador (Cabanillas-Terán et al. 2016).

In conjunction with previous findings of partitioning of shelter resources between *P. guttatus* and *P. argus* in the reef habitat (Sharp et al. 1997, Acosta and Robertson 2003, Lozano-Álvarez et al. 2007), our results also provide insight into the differences in traits that underlie niche differentiation between these sympatric species (reviewed in Briones-Fourzán 2014) and support the niche theory prediction that species-specific specialization in resource use is a primary mechanism allowing the stable coexistence among competing species within a local community (Chesson 2000, Kylafis and Loreau 2011). Further studies combining SIA with fatty acid analyses (e.g. Hooker et al. 2001, Guest et al. 2009, Kelly and Scheibling 2012) or genetic analyses (O’Roarke et al. 2012, Pompanon et al. 2012) could provide more details about the carbon sources and prey organisms consumed by these and other sympatric congeneric lobsters.

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